

Hierarchical Clustering

- Broadly categorizing, there are two ways of performing Hierarchical Clustering.

1. Agglomerative Clustering:

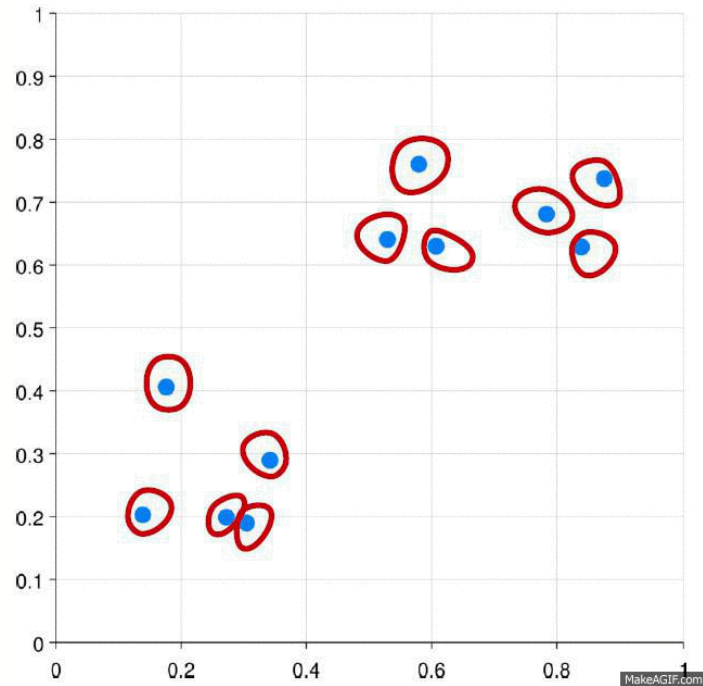
- The word agglomerative suggests combining things
- It is a bottom-up approach
- Agglomerative clustering starts with the assumption that every data point is a cluster
- Then, it groups the clusters that are close to each other until there is only a single cluster left

2. Divisive Clustering:

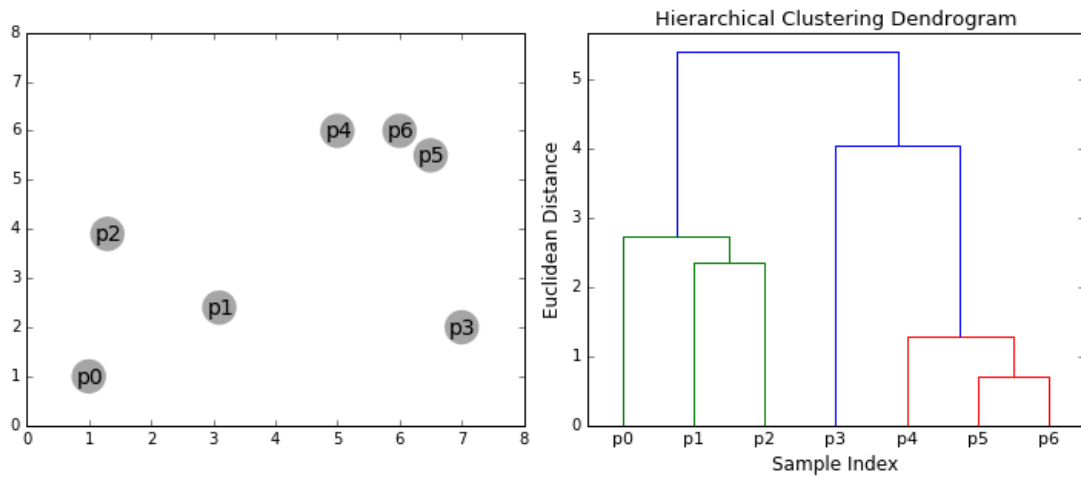
- It is the complete opposite of the agglomerative approach
- It is a top-down approach
- It starts with one big cluster that contains all the data points.
- It then divides the points into different clusters till each data point is a cluster itself

Agglomerative Clustering

- The steps involved in Agglomerative Clustering are:
 1. Assume each point is a cluster (n datapoints -> n clusters)
 2. Compute Proximity Matrix ($P_{n \times n}$)
 3. Repeat until a single cluster is left:
 - a. Merge the closest clusters
 - b. Update the proximity matrix



- If we visualize this, this looks like a Tree, but there is another name that is often used in Data Mining terminology which is called Dendrogram.



Proximity Matrix

- A proximity matrix is a matrix of distances or similarity.
 - The word proximity suggests how close things are
- Say, at any point we're having C_m clusters. For each of the pairs of clusters, the proximity matrix P will indicate the similarity between clusters C_i and C_j .
- Initially, the proximity matrix P will be $N*N$ matrix.
- Suppose cluster C_i and C_j , where $i \neq j$, are similar and they have the smallest value in the proximity matrix, then those clusters will be combined and the proximity matrix will get updated
- The new matrix will be a $N-1*N-1$ matrix, as two clusters have combined.
- One can use the following distances for computing the values of proximity matrices.
 1. Using Euclidean Distance between the centroids of two clusters C_i and C_j .
 2. Maximum distance between two points x_i and x_j , such that $x_i \in C_i$ and $x_j \in C_j$.
 3. The minimum distance between two points x_i and x_j , such that $x_i \in C_i$ and $x_j \in C_j$.

4. Average
$$\sum_{x_i \in C_i} \sum_{x_j \in C_j} \frac{\text{dist}(x_i x_j)}{|C_i||C_j|} \quad \text{Distance:}$$

5. Ward's
$$\sum_{x_i \in C_i} \sum_{x_j \in C_j} \frac{\text{dist}(x_i x_j)^2}{|C_i||C_j|} \quad \text{Distance:}$$

Limitations of Hierarchical Clustering

1. With large datasets, Agglomerative Clustering does not work well
 - a. Space Complexity = $O(n)$: Proximity Matrix
 - b. Time Complexity = $O(n^2)$
2. Unlike K-means where we try to minimize **within-cluster distance**, there is **no mathematical objective** that is being minimized in Agglomerative clustering.